

Chiharu Mamiya

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Professional Summary

Electrical Engineering student with robust experience in software development, hardware design, and systems engineering. Skilled in Python, C/C++, VHDL/Verilog, FPGA, and embedded systems. Proven track record optimizing power system tools, automating testing, and deploying ML models for IoT and biomedical applications. Strong leadership in managing cross-functional teams and projects, with international research and internship experience. Committed to delivering reliable, efficient technical solutions.

Engineering Experience

Technical Consultant Intern Siemens *Schenectady, NY* **06/2025 - Present**

- Validated grid models and edited ACCC Python tools to streamline contingency and power flow analysis.
- Improved TSAT plotting tool runtime by 94% and supported flow gate upgrades to enhance simulation performance.
- Built test suites and flowcharts to aid debugging, regression testing, and onboarding, improving code reliability.

Hardware Engineering Intern FH Münster *Münster, Germany* **09/2024 – 11/2024**

- Designed and tested schematics on Red Pitaya boards, improving quantum sensing FPGA timing performance by 10%.
- Debugged hardware using VHDL/Verilog in a Linux environment, reducing signal errors, and increasing system uptime.

Software Developer Intern Siemens *Schenectady, NY* **06/2024 - 08/2024**

- Automated XML data extraction via Python scripts, reducing manual workload by 25% and pushing updates to GitHub.
- Migrated legacy C++ modules to Fortran and MFC, enhancing cross-platform integration across x86-based infrastructure.
- Communicated progress in daily stand-ups, contributing to 2-week early project delivery.

IT Intern Grew Bancroft *Remote* **04/2024 – 08/2024**

- Enhanced React/Node.js application frontend, reducing page load times by 25% and improving UI responsiveness.
- Automated user testing with Jest, streamlining QA workflows and reducing manual testing by 20%.

IoT Student Researcher Union College *Schenectady, NY* **10/2023 – 08/2024**

- Processed and labeled 1,200+ motion datasets under IRB protocols, supporting ethical sports biomechanics research.
- Trained SVM/KNN models in Python, achieving 80% classification accuracy for expert vs. novice athlete skills.
- Co-authored paper accepted by IEEE AIIoT; collaborated with 4 professors and 1 peer to iterate ML and experimental design.

Systems Engineering Intern Medtronic *Fridley, MN* **06/2023 – 08/2023**

- Developed Python system testing automation scripts with Appium and ADB, boosting embedded system testing efficiency by 30%.
- Conducted system tests on a closed-loop brain stimulation device using oscilloscopes and function generators.
- Presented findings in weekly meetings, helping resolve hardware-software issues faster.

Leadership & Management Experience

PR Intern Grew Bancroft *Remote* **12/2021 – 08/2024**

- Led a cross-functional team across 3+ time zones using Discord and bi-weekly meetings, increasing efficiency, and cutting delays by 25%.
- Negotiated with 5+ vendors and adjusted project timelines, ensuring 100% on-time delivery of events despite scheduling conflicts.
- Conducted monthly engagement reviews and implemented changes, increasing program applications by 30%.

Project Management Intern PivotGen *Remote* **01/2024 - 06/2024**

- Facilitated communication between 10+ engineers using Microsoft Teams and shared docs, accelerating issue resolution by 15%.
- Managed Gantt charts and tracked progress of 96 turbines, improving resource allocation efficiency by 20%.
- Performed bi-weekly resource audits to ensure availability, preventing delays and maintaining a 100% on-time material delivery rate.

House Coordinator Union Housing Office *Schenectady, NY* **09/2023 - 06/2024**

- Managed a \$16,000 budget to execute 6+ large-scale campus events with 500+ attendees, coordinating with dining and student boards.
- Coordinated with 10 event staffs via Slack and weekly meetings, resolving incidents promptly and ensuring seamless event operations.
- Built a Google Sheets tracking system for incidents and maintenance, reducing response times by 40%.

President Union Neurotech *Schenectady, NY* **10/2021 - 06/2024**

- Directed a 6-member team to develop an EEG-based drone control system, achieving 95% accuracy using Python and ML tools.
- Hosted 2–3 neurotech workshops each term, maintaining consistent engagement from 40+ participants per event.
- Led a brainwave-art project showcased at BR41N.IO, earning 4th place out of 20+ international student teams.

Projects

Capstone Project: Smart Microgrid Node, Senior Thesis *Capstone Project*04/2024 - Present

- Developing a solar-powered, scalable, and cost-effective microgrid node integrating IoT sensors and STM32 microcontrollers.
- Designing hardware and firmware for real-time energy monitoring, load balancing, and peer-to-peer communication.
- Implementing communication protocols for decentralized energy data exchange and optimization using Python.

IoT and Gesture Recognition System with Neural Network Deployment on ESP32, *Internet of Things*05/2024 - 06/2024

- Developed an agriculture IoT platform with Bluetooth, WiFi, and AWS Core, increasing sensor range by 50 meters.
- Achieved 96% gesture recognition accuracy using TensorFlow and MPU6050, validated across 4 gestures.
- Deployed real-time control on ESP32 boards, enabling live RC car navigation via remote control gestures.

Other Experience

Peer Career AdvisorUnion Career CenterSchenectady, NY03/2023 - 04/2024

- Guided 100+ students in career exploration, networking, and resumes through 1-on-1 advising, resulting in 80% internship placements.
- Developed Google Sheets tracking systems for networking & internship applications, improving student engagement by 40%.
- Completed 20+ hours of career counseling training and presented findings on AI enhanced advising to senior staffs.

Market ResearcherAlbany Med Health SystemAlbany, NY01/2024 - 06/2024

- Coordinated timelines for a 6-month research initiative using Microsoft Project, increasing milestone completion rate to 90%.
- Conducted 50+ market research interviews and analyzed data in Excel, contributing to a 15% improvement in pitch effectiveness.
- Balanced research, data synthesis, and preparation development, helping the team meet 100% of internal deadlines.

Program AnalystNASA L'SPACE Mission Concept AcademyRemote09/2024 – 12/2024

- Managed \$425M lunar mission concept, optimizing budget across instruments, operations, and a dual-rover system.
- Built a traceability matrix linking mission goals to outcomes, guiding trade studies and cost-performance decisions.
- Collaborated with 10+ students across 10 institutions to define requirements and assess 4 lunar landing sites.

Skills

- Python, C, C++, VHDL, Verilog, MATLAB, Simulink, TypeScript, Git, Visual Studio, Quartus, Vivado, Red Pitaya, Appium, Jest, FPGA, ARM CPUs, IoT Core, STM32, ESP32, DE10, PyTorch, TensorFlow, TinyML, SVM, KNN, Gantt Charts, Agile, Microsoft Project, API Integration, Data Collection & Analysis, Signal Analysis, System Testing

Education

Bachelor of ScienceUnion CollegeSchenectady, NY

- Major in Electrical Engineering, Minor in Biomedical Engineering, GPA: 3.86
- Honors: Tau Beta Pi Engineering Honor Society, JASSO Scholar, Grew Bancroft Scholar UAS7 Scholar, NYSTAR Scholar
- Coursework: Digital Design, Internet of Things, Embedded Microcontrollers, Control Systems, Semiconductor Systems